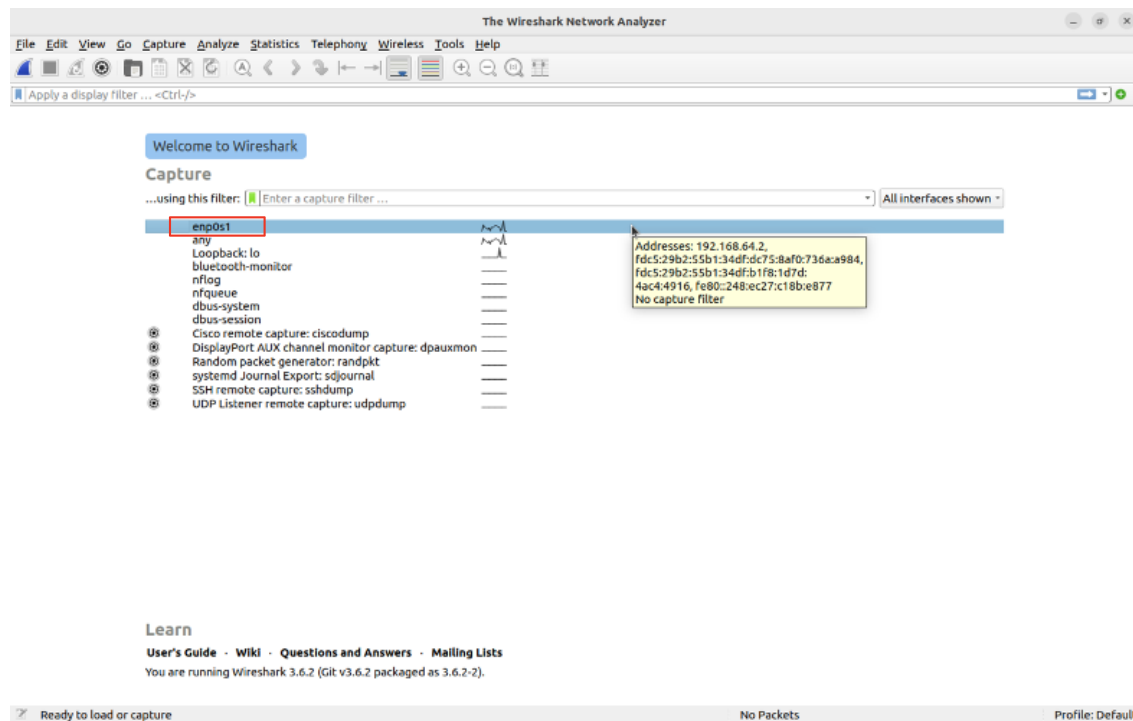


Question number 1:

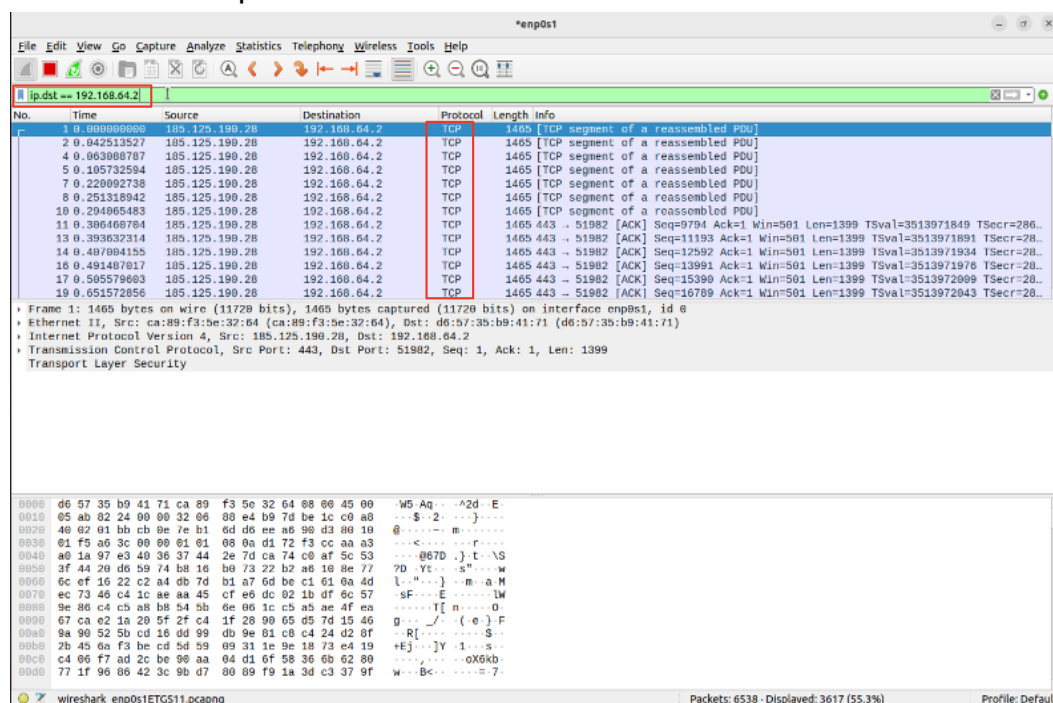


As one can see we have here opened wireshark as required, and we have our pointer on the interface that's connected to the external internet.

Question number 2:

A:

we shall filter the packets based on the ip destination based on the address 192.168.64.2 (Our chosen ip destination) with the correct search term in the command line: `ip.dst == 192.168.64.2`



B:

filtering the packets by source port 80. one can clearly see in the screenshot the protocol that is being used is tcp.

Filter: tcp.port == 80 || udp.port == 80

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	185.125.190.28	192.168.64.2	TCP	1465	[TCP segment of a reassembled PDU]
2	0.042513527	185.125.190.28	192.168.64.2	TCP	1465	[TCP segment of a reassembled PDU]
3	0.042534998	192.168.64.2	185.125.190.28	TCP	66	51982 → 443 [ACK] Seq=1 Ack=2799 Win=4689 Len=0 TSval=2862850830 TSecr=35139...
4	0.063080787	185.125.190.28	192.168.64.2	TCP	1465	[TCP segment of a reassembled PDU]
5	0.105732594	185.125.190.28	192.168.64.2	TCP	1465	[TCP segment of a reassembled PDU]
6	0.105753190	192.168.64.2	185.125.190.28	TCP	66	51982 → 443 [ACK] Seq=1 Ack=5597 Win=4689 Len=0 TSval=2862850893 TSecr=35139...
7	0.270692730	185.125.190.28	192.168.64.2	TCP	1465	[TCP segment of a reassembled PDU]
8	0.251318942	185.125.190.28	192.168.64.2	TCP	1465	[TCP segment of a reassembled PDU]
9	0.251337459	192.168.64.2	185.125.190.28	TCP	66	51982 → 443 [ACK] Seq=1 Ack=8395 Win=4689 Len=0 TSval=2862851038 TSecr=35139...
10	0.294065483	185.125.190.28	192.168.64.2	TCP	1465	[TCP segment of a reassembled PDU]
11	0.306469704	185.125.190.28	192.168.64.2	TCP	1465	443 → 51982 [ACK] Seq=9794 Ack=1 Win=501 Len=1399 TSval=3513971849 TSecr=286...
12	0.306473952	192.168.64.2	185.125.190.28	TCP	66	51982 → 443 [ACK] Seq=1 Ack=11193 Win=4689 Len=0 TSval=2862851094 TSecr=3513...
13	0.393632314	185.125.190.28	192.168.64.2	TCP	1465	443 → 51982 [ACK] Seq=11193 Ack=1 Win=501 Len=1399 TSval=3513971891 TSecr=28...

Frame 11: 1465 bytes on wire (11720 bits), 1465 bytes captured (11720 bits) on interface enp0s1, id 0
Ethernet II, Src: ca:89:f3:5e:32:64 (ca:89:f3:5e:32:64), Dst: d6:57:35:b0:41:71 (d6:57:35:b0:41:71)
Internet Protocol Version 4, Src: 185.125.190.28, Dst: 192.168.64.2
Transmission Control Protocol, Src Port: 443, Dst Port: 51982, Seq: 1, Ack: 1, Len: 1399
Transport Layer Security

Packets: 18750 · Displayed: 18615 (99.3%) Profile: Default

C:

Filtering for UDP packets

Filter: udp!

No.	Time	Source	Destination	Protocol	Length	Info
7	0.110497057	192.168.64.1	224.0.0.251	NDNS	685	Standard query 0x0000 PTR lb.dns-sd.udp.local, "Q" question PTR _airport...
8	3.110497065	fe80::c899:f3ff:fe5...	ff02::fb	NDNS	765	Standard query 0x0000 PTR lb.dns-sd.udp.local, "Q" question PTR _airport...
9	54.37.042012386	192.168.64.2	192.168.64.1	DNS	95	Standard query 0xd8e0 A normandy.cdn.mozilla.net OPT
10	55.37.042139134	192.168.64.2	192.168.64.1	DNS	95	Standard query 0x1a7c AAAA normandy.cdn.mozilla.net OPT
11	56.37.057750136	192.168.64.1	192.168.64.2	DNS	158	Standard query response 0xd8e0 A normandy.cdn.mozilla.net CNAME normandy-cdn...
12	57.37.067064343	192.168.64.1	192.168.64.2	DNS	228	Standard query response 0x1a7c AAAA normandy.cdn.mozilla.net CNAME normandy...
13	58.37.067322793	192.168.64.2	192.168.64.1	DNS	184	Standard query 0x4dc2 AAAA normandy-cdn.services.mozilla.com OPT
14	59.37.068558054	192.168.64.1	192.168.64.2	DNS	185	Standard query response 0x49c2 AAAA normandy-cdn.services.mozilla.com SOA ns...
15	73.37.100759488	192.168.64.2	192.168.64.1	DNS	85	Standard query 0xe8b9 A r3.o.lencr.org OPT
16	74.37.100796583	192.168.64.2	192.168.64.1	DNS	85	Standard query 0x1483 AAAA r3.o.lencr.org OPT
17	75.37.116171636	192.168.64.1	192.168.64.2	DNS	184	Standard query response 0xe8b9 A r3.o.lencr.org CNAME o.lencr.edgesuite.net
18	76.37.116171928	192.168.64.1	192.168.64.2	DNS	288	Standard query response 0x1483 AAAA r3.o.lencr.org CNAME o.lencr.edgesuite.n...
19	99.37.161785392	192.168.64.2	192.168.64.1	DNS	107	Standard query 0xf4c3 A classiffy-client.services.mozilla.com OPT

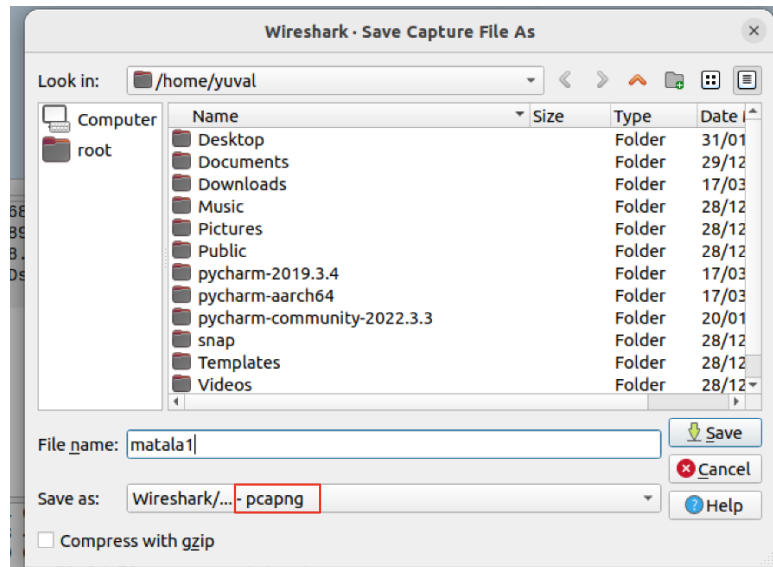
Frame 7: 685 bytes on wire (5480 bits), 685 bytes captured (5480 bits) on interface enp0s1, id 0
Ethernet II, Src: ca:89:f3:5e:32:64 (ca:89:f3:5e:32:64), Dst: IPv4mcast_fb (01:00:5e:00:00:fb)
Internet Protocol Version 4, Src: 192.168.64.1, Dst: 224.0.0.251
User Datagram Protocol, Src Port: 5353, Dst Port: 5353
Multicast Domain Name System (query)

Packets: 2561 · Displayed: 150 (5.9%) Profile: Default

Question number 3:

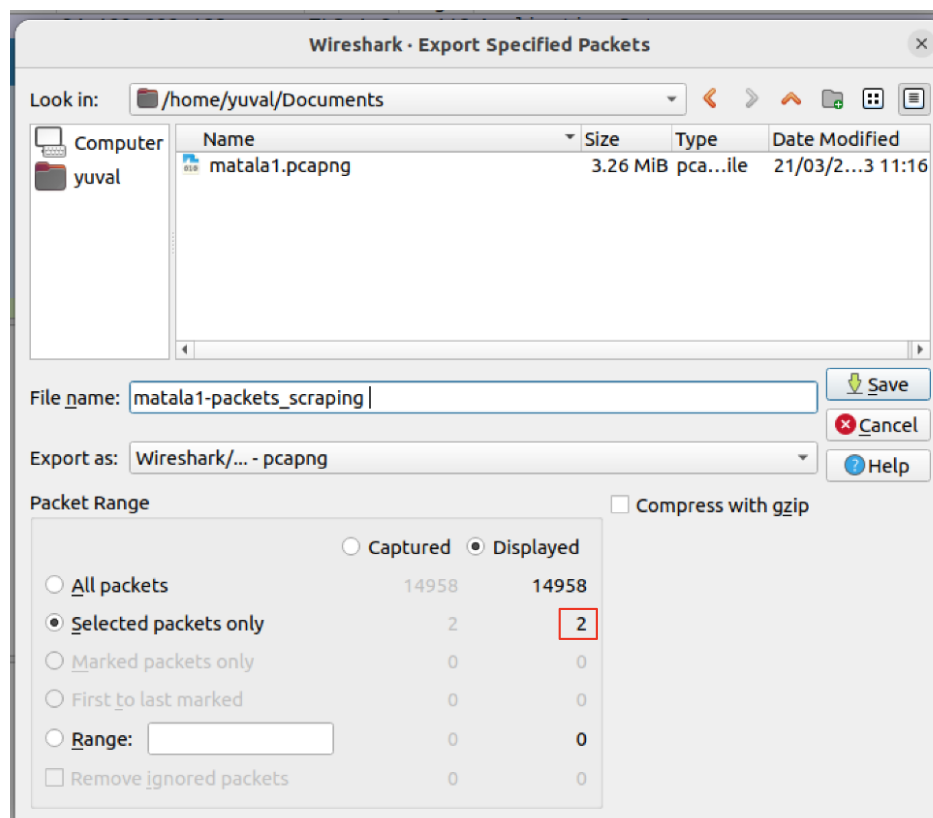
We saved the packets in pcapng format.

.pcapng is one of the appropriate formats for saving a recording of a wireshark capture.

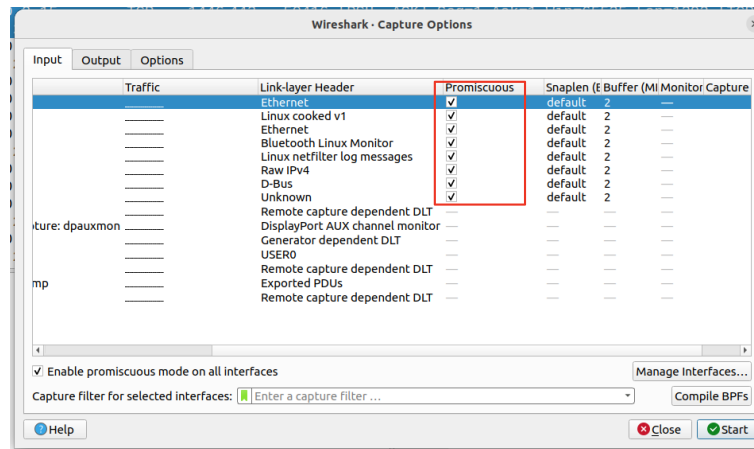


Question number 4:

As one can clearly see here, we selected 2 packets by using the shift button and selecting the packets we wanted. we can the export the specified packets that we chose.



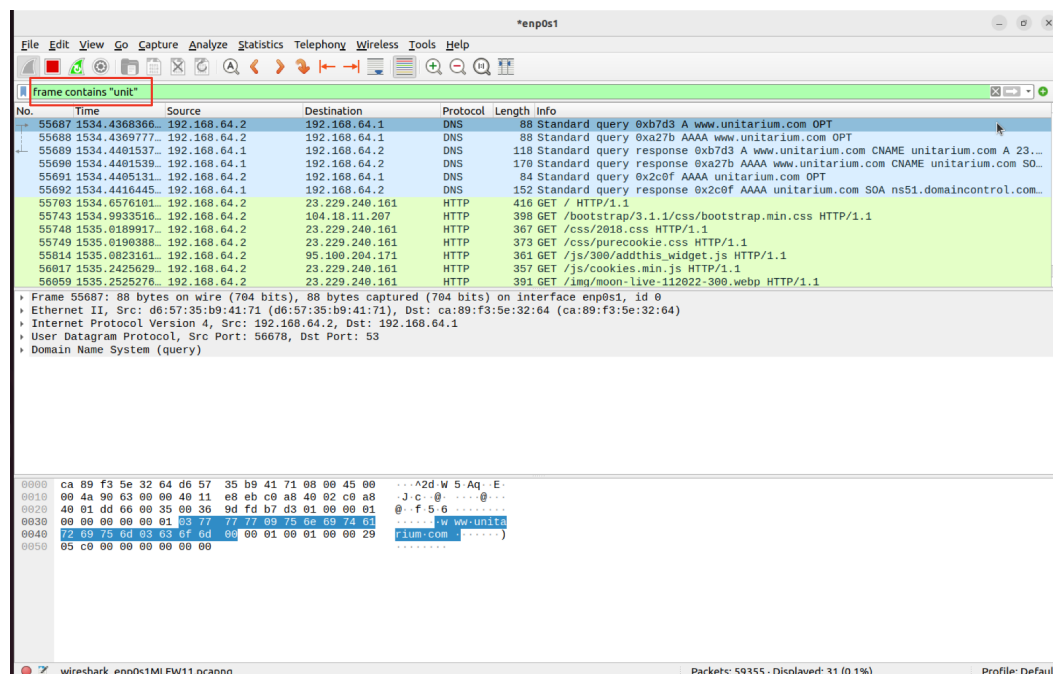
Question number 5:



As one can see we have located the promiscuous mode in the appropriate settings and checkmarked them.

- This mode lets all network data packets be accessed and viewed by all network adapters with this mode enabled.
- This mode is used and implemented by a program that captures all network traffic visible on all configured network adapters on a system. Unlike the regular mode, in which the sniffer filters from and to the computer its running on, in promiscuous mode, all traffic will be passed.
- this can be used in the domains of network security, monitoring and administration

Question number 6:



The Wireshark filter filters for frames containing the string "unit", thus as a result because we entered the website stated in the question that clearly includes the string "unit" in the URL, we receive filtered results containing that website.

HTTP:

שאלה מספר 7:

The screenshot shows a Wireshark packet capture on interface enp0s1. The packet list table is as follows:

No.	Time	Source	Destination	Protocol	Length	Info
61118	1978.2352777...	fd5:29b2:55b1:34df...	2620:2d:4000:1::2a	HTTP	173	GET / HTTP/1.1
61119	1978.3133117...	2620:2d:4000:1::2a	fd5:29b2:55b1:34df...	HTTP	233	HTTP/1.1 204 No Content
61173	2007.3452169...	192.168.64.2	192.229.221.95	OCSP	491	Request
61175	2007.4248677...	192.229.221.95	192.168.64.2	OCSP	807	Response
61225	2012.9428277...	192.168.64.2	172.217.22.99	OCSP	493	Request
61227	2013.2574371...	172.217.22.99	192.168.64.2	OCSP	767	Response
61313	2083.7460297...	192.168.64.2	91.189.91.48	HTTP	153	GET / HTTP/1.1
61314	2083.9039561...	91.189.91.48	192.168.64.2	HTTP	213	HTTP/1.1 204 No Content
61492	2104.5778240...	192.168.64.2	94.23.157.180	HTTP	418	GET / HTTP/1.1
61498	2104.6625189...	94.23.157.180	192.168.64.2	HTTP	470	HTTP/1.1 200 OK (text/html)
61519	2104.8540585...	192.168.64.2	52.217.203.88	HTTP	372	GET /cc.silktide.com/cookieconsent.latest.min.js HTTP/1.1
61532	2105.0031668...	52.217.203.88	192.168.64.2	HTTP/X..	308	HTTP/1.1 403 Forbidden

The details pane for the selected packet (61492) shows the following structure:

- Frame 61492: 418 bytes on wire (3344 bits), 418 bytes captured (3344 bits) on interface enp0s1, id 0
- Ethernet II, Src: d6:57:35:b9:41:71 (d6:57:35:b9:41:71), Dst: ca:89:f3:5e:32:64 (ca:89:f3:5e:32:64)
- Internet Protocol Version 4, Src: 192.168.64.2, Dst: 94.23.157.180
- Transmission Control Protocol, Src Port: 46276, Dst Port: 80, Seq: 1, Ack: 1, Len: 352
- Hypertext Transfer Protocol**
 - GET / HTTP/1.1\r\n
 - Host: www.shai-online.com\r\n
 - User-Agent: Mozilla/5.0 (X11; Ubuntu; Linux aarch64; rv:109.0) Gecko/20100101 Firefox/109.0\r\n
 - Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,*/*;q=0.8\r\n
 - Accept-Language: en-US,en;q=0.5\r\n
 - Accept-Encoding: gzip, deflate\r\n
 - Connection: keep-alive\r\n
 - Upgrade-Insecure-Requests: 1\r\n
 - \r\n

The packet bytes pane shows the raw data of the request, including the GET method and host information.

we can identify the queries by the GET method.

In the screenshot we can deduce the request from my local IP and the ones from sha1 (des-94.23.157.180).

if we click on the packets we can clearly see the area of the hypertext and expand the contents and see the name of the host. Furthermore one can see in the screenshot that before a request there is a right pointing triangular arrow , the answer to the request is with a left pointed triangular arrow.

Question number 8:

A:

The screenshot shows a Wireshark packet capture on interface enp0s1. The packet list table is as follows:

No.	Time	Source	Destination	Protocol	Length	Info
61118	1978.2352777...	fd5:29b2:55b1:34df...	2620:2d:4000:1::2a	HTTP	173	GET / HTTP/1.1
61119	1978.3133117...	2620:2d:4000:1::2a	fd5:29b2:55b1:34df...	HTTP	233	HTTP/1.1 204 No Content
61173	2007.3452169...	192.168.64.2	192.229.221.95	OCSP	491	Request
61175	2007.4248677...	192.229.221.95	192.168.64.2	OCSP	807	Response
61225	2012.9428277...	192.168.64.2	172.217.22.99	OCSP	493	Request
61227	2013.2574371...	172.217.22.99	192.168.64.2	OCSP	767	Response
61313	2083.7460297...	192.168.64.2	91.189.91.48	HTTP	153	GET / HTTP/1.1
61314	2083.9039561...	91.189.91.48	192.168.64.2	HTTP	213	HTTP/1.1 204 No Content
61492	2104.5778240...	192.168.64.2	94.23.157.180	HTTP	418	GET / HTTP/1.1
61498	2104.6625189...	94.23.157.180	192.168.64.2	HTTP	470	HTTP/1.1 200 OK (text/html)
61519	2104.8540585...	192.168.64.2	52.217.203.88	HTTP	372	GET /cc.silktide.com/cookieconsent.latest.min.js HTTP/1.1
61532	2105.0031668...	52.217.203.88	192.168.64.2	HTTP/X..	308	HTTP/1.1 403 Forbidden

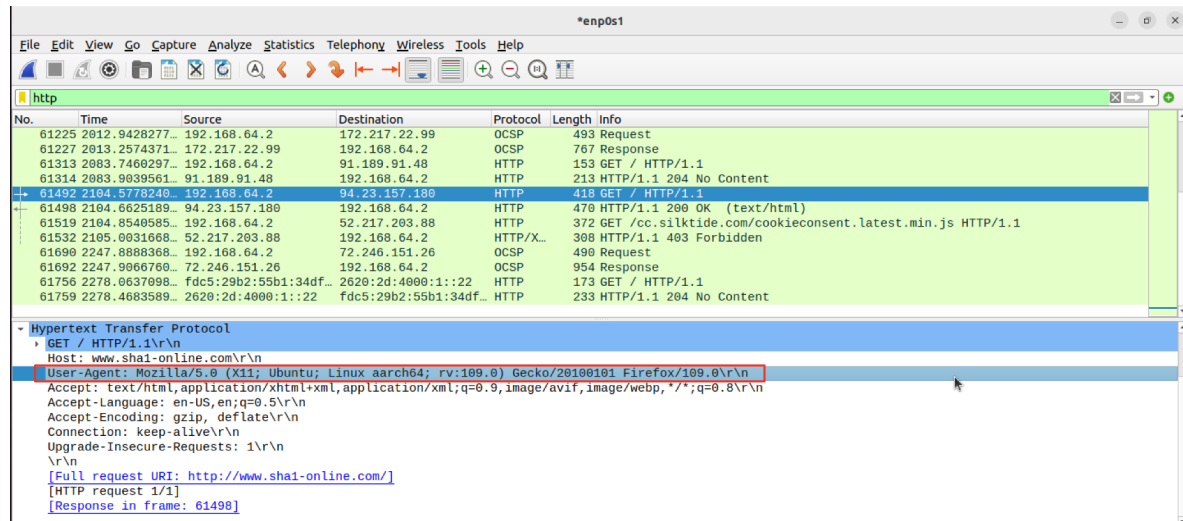
The details pane for the selected packet (61492) shows the following structure:

- Frame 61492: 418 bytes on wire (3344 bits), 418 bytes captured (3344 bits) on interface enp0s1, id 0
- Ethernet II, Src: d6:57:35:b9:41:71 (d6:57:35:b9:41:71), Dst: ca:89:f3:5e:32:64 (ca:89:f3:5e:32:64)
- Internet Protocol Version 4, Src: 192.168.64.2, Dst: 94.23.157.180
- Transmission Control Protocol, Src Port: 46276, Dst Port: 80, Seq: 1, Ack: 1, Len: 352
- Hypertext Transfer Protocol**
 - GET / HTTP/1.1\r\n
 - Host: www.shai-online.com\r\n
 - User-Agent: Mozilla/5.0 (X11; Ubuntu; Linux aarch64; rv:109.0) Gecko/20100101 Firefox/109.0\r\n
 - Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,*/*;q=0.8\r\n
 - Accept-Language: en-US,en;q=0.5\r\n
 - Accept-Encoding: gzip, deflate\r\n
 - Connection: keep-alive\r\n
 - Upgrade-Insecure-Requests: 1\r\n
 - \r\n

The packet bytes pane shows the raw data of the request, including the GET method and host information.

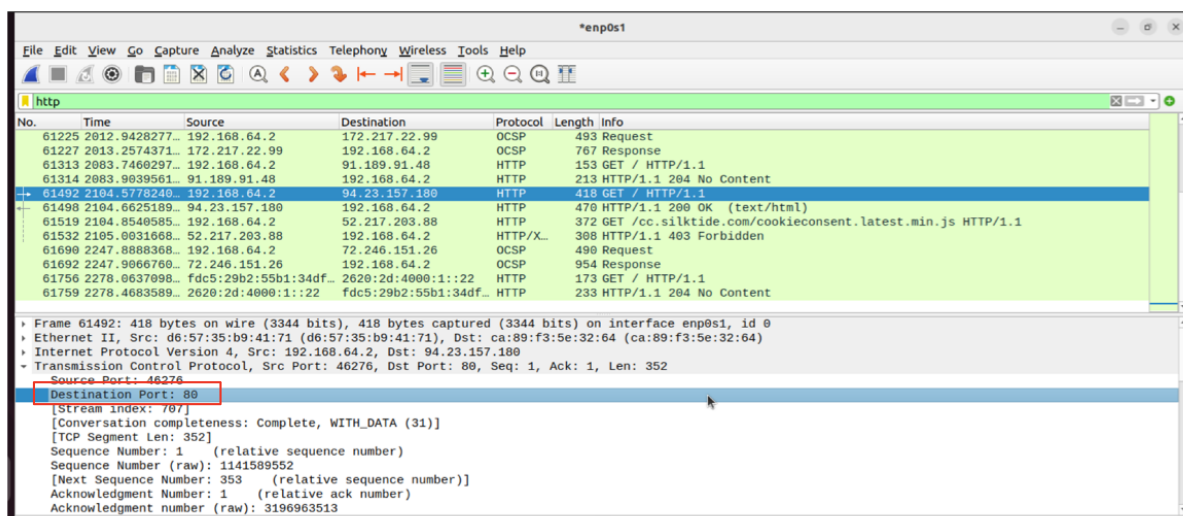
the HTTP version is stated after the GET request and is version 1.1

B:



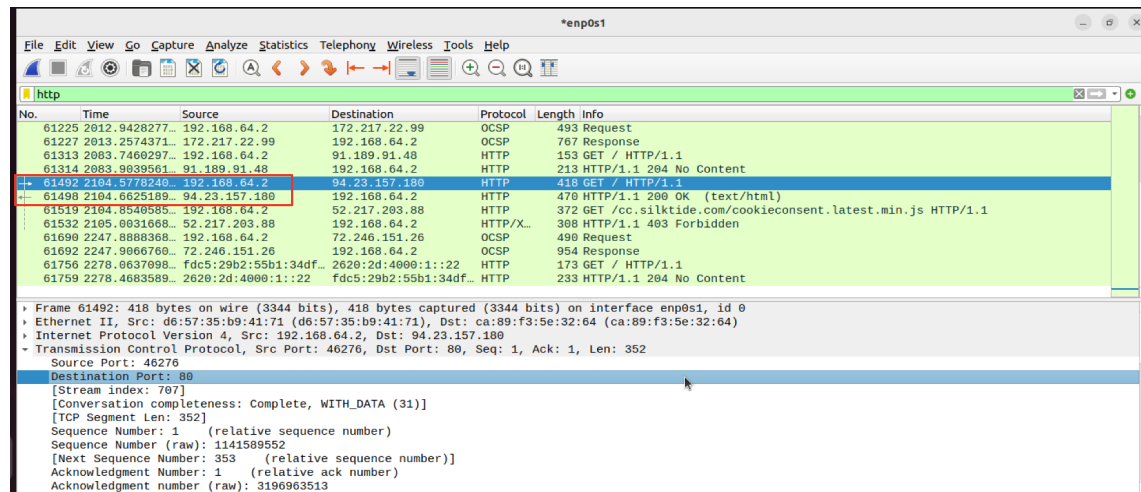
by the stated user agent we can see the request was made by ubuntu linux through firefox

C:



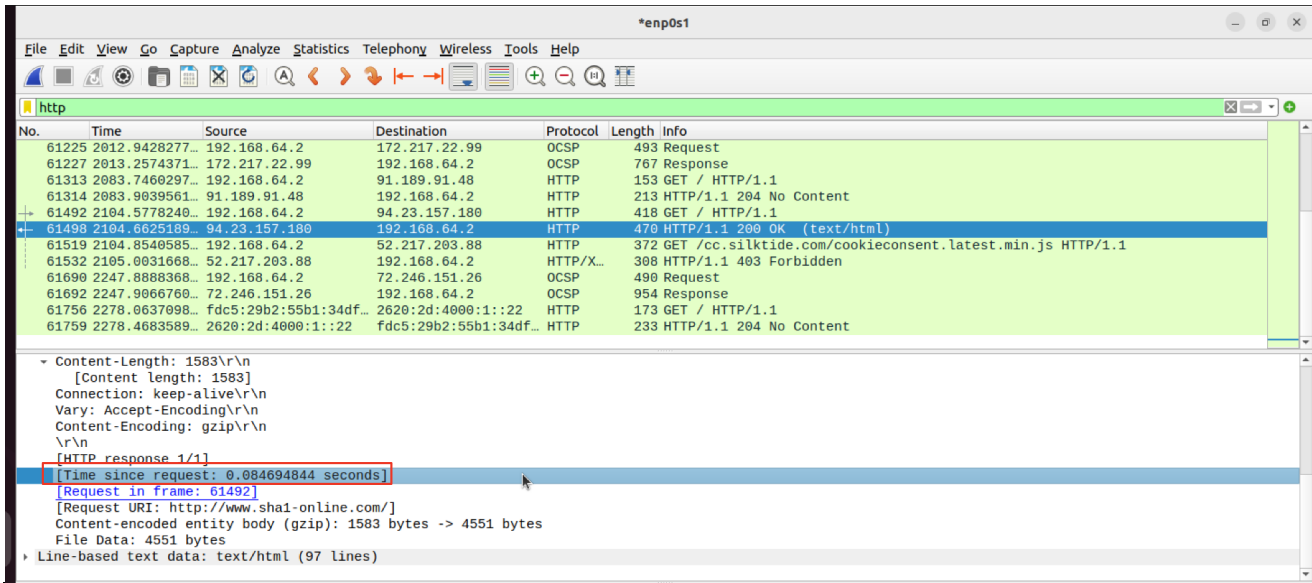
The destination port is port 80. this is the standard port for HTTP

D:



The request response package can be seen under the original request.
Furthermore, we can identify the response package by the left arrow.

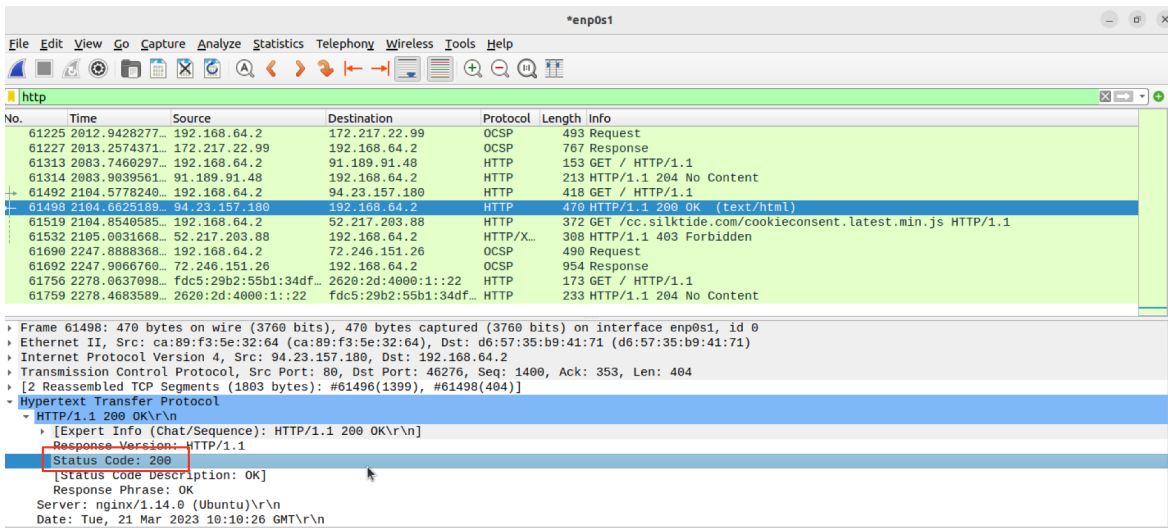
E:



0.84694844 seconds.

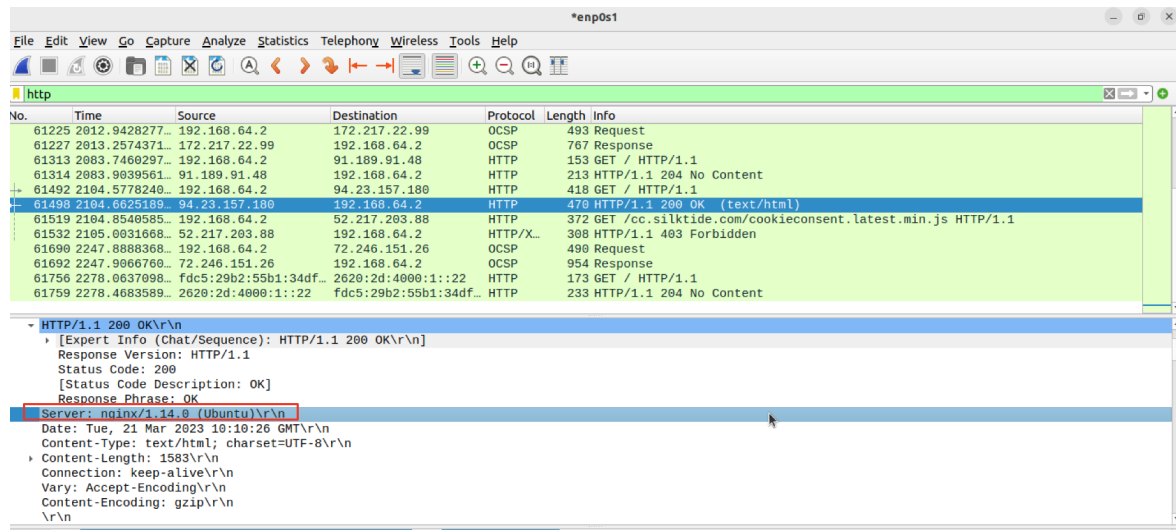
QUESTION NUMBER 9:

A:



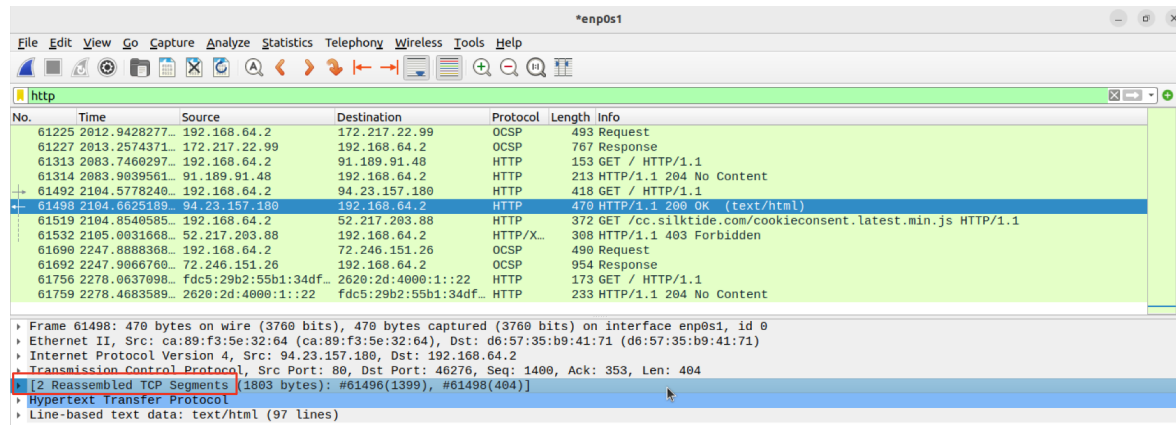
status code = 200.

B:



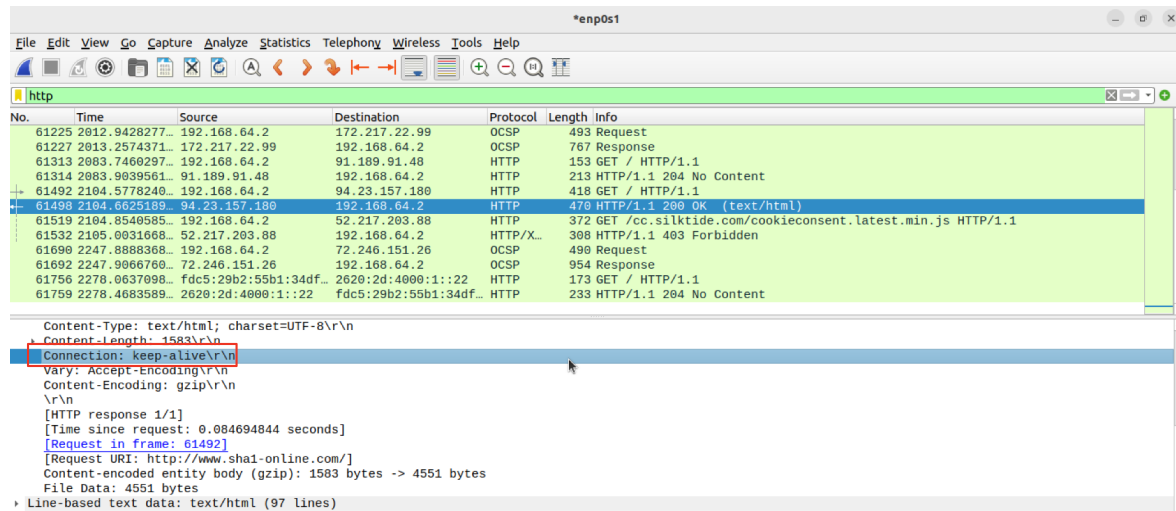
the server: nginx/ 1.14.8 (ubuntu)

C:



2 packets.

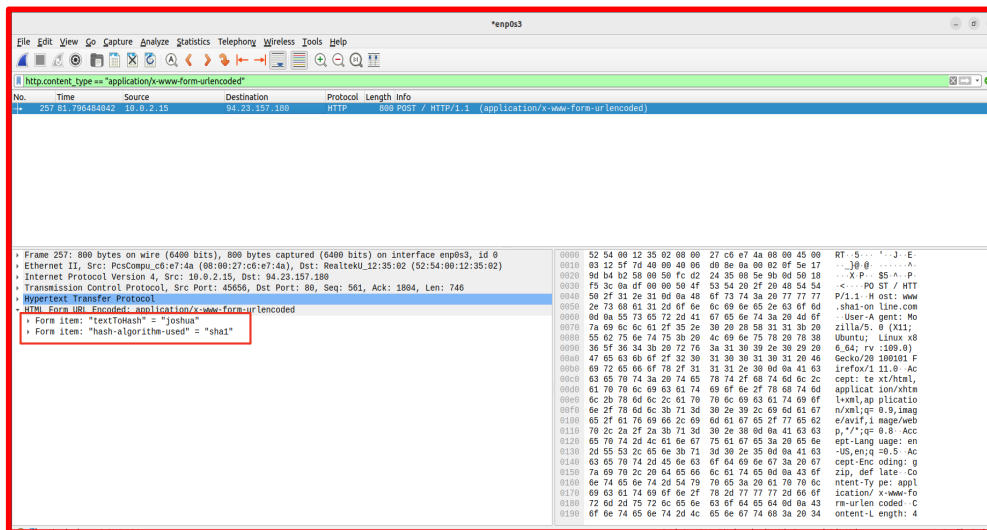
D:



keep-alive connection means keep the connection “alive” and open. This means we don't need to connect again and again everytime we connect with the website.

QUESTION NUMBER 10:

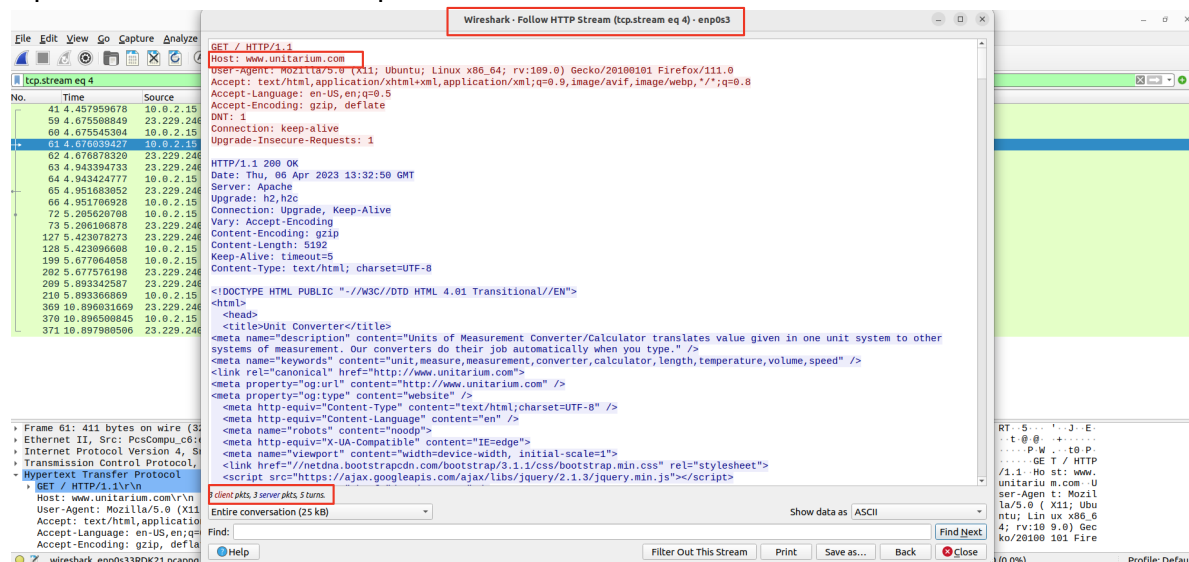
- 1) The computation was computed on the server side. we come to the conclusion as the protocol was OSCP , this basically means a certificate indicating the success of the operation on the server side.
- 2) The sent parameters are “Joshua”, as parameter to hash, and “sha1”, as the hashing method.



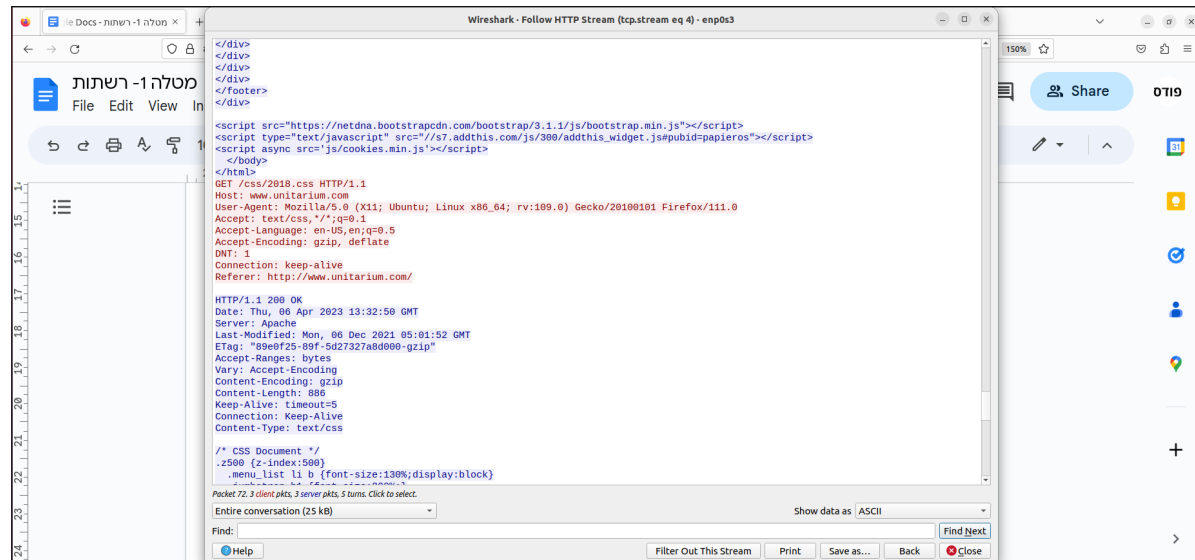
- 3) A reason this computation might have been done on the server instead of the browser is the computational power needed to complete the sha1 algorithm is intense and needs a large amount of computing power.
- 4) One danger could be that “sniffers” could intercept the data being sent back. if the answer was done in the browser no sniffers would intercept the answer.

Question number 11:

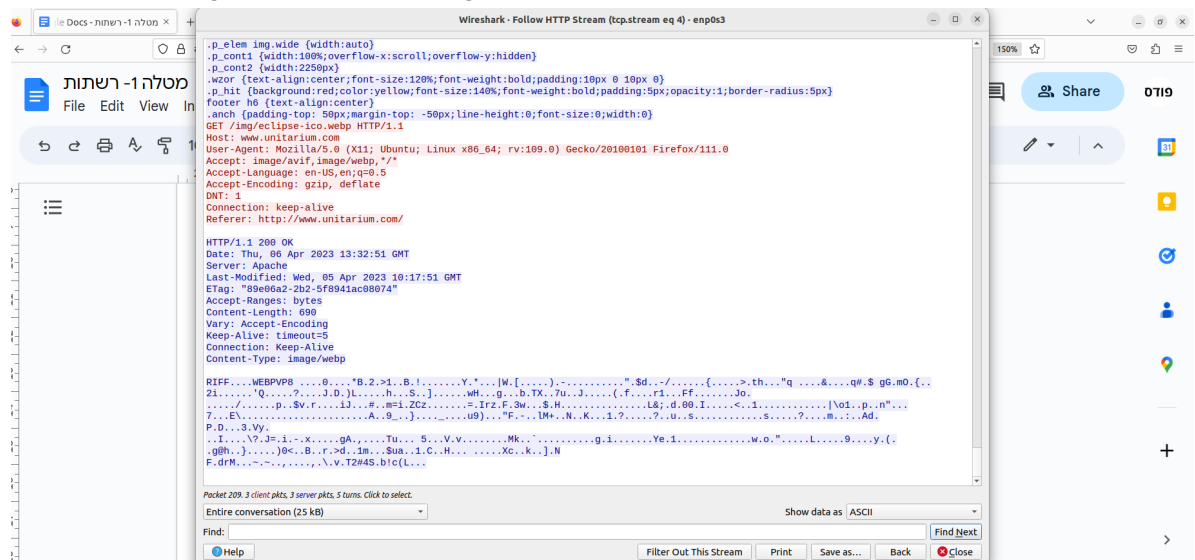
3 packets from client and 3 packets from server 3 were returned from the server.



the first packet contained the html document as shown above
the second package contained the css (the file that decorates the html file)



the third package contained a image in ico.webp format



Question number 12:

We entered amazon.com

then there were multiple DNS queries trying to find amazons ip address (trying in local catch and then searching local and wider networks). once found there was an http get request with a return value of moved permanently. then there were other dns queries sent until we finally got the page back and loaded. (it looked like amazon was registering a key for our new entry for our session).

DNS

Question number 13:

A:

no this is not an authoritative server. the server is 127.0.0.00 this is our local server as we are working on a virtual machine this answer came up.

```
jodogo@jodogo-VirtualBox: ~  
jodogo@jodogo-VirtualBox:~$ nslookup cats.com  
Server:      127.0.0.53  
Address:     127.0.0.53#53  
  
Non-authoritative answer:  
Name:   cats.com  
Address: 104.22.72.237  
Name:   cats.com  
Address: 104.22.73.237  
Name:   cats.com  
Address: 172.67.13.90
```

B:

the three ip address's for the domain are:

104.22.72.235
104.22.73.237
172.67.13.90

Question number 14:

A:

4 queries

The image shows a Wireshark packet capture of DNS traffic. The packet list pane displays four queries (159-163) for the domain cats.com. The packet details pane for packet 159 shows the following structure:

- Frame 159: 127 bytes on wire (1016 bits), 127 bytes captured (1016 bits) on interface enp0s3, id 0
- Ethernet II, Src: RealtekU_12:35:02 (52:54:00:12:35:02), Dst: PcsCompu_c6:e7:4a (08:00:27:c6:e7:4a)
- Internet Protocol Version 4, Src: 185.180.100.65, Dst: 10.0.2.15
- User Datagram Protocol, Src Port: 53, Dst Port: 59917
- Domain Name System (response)
 - Transaction ID: 0xe174
 - Flags: 0xb180 Standard query response, No error
 - Questions: 1
 - Answer RRs: 3
 - Authority RRs: 0
 - Additional RRs: 1
 - Queries
 - cats.com: type A, class IN
 - Name: cats.com
 - [Name Length: 8]
 - [Label Count: 2]
 - Type: A (Host Address) (1)
 - Class: IN (0x0001)
 - Answers
 - cats.com: type A, class IN, addr 104.22.73.237
 - Name: cats.com
 - Class: IN (0x0001)

B:

the first 3 are get the response of each of the ip addresses, the last one get a general response for all 3 ip address. The queries are A query, meaning a question or a request for information expressed in a formal manner.

C:
port 53:

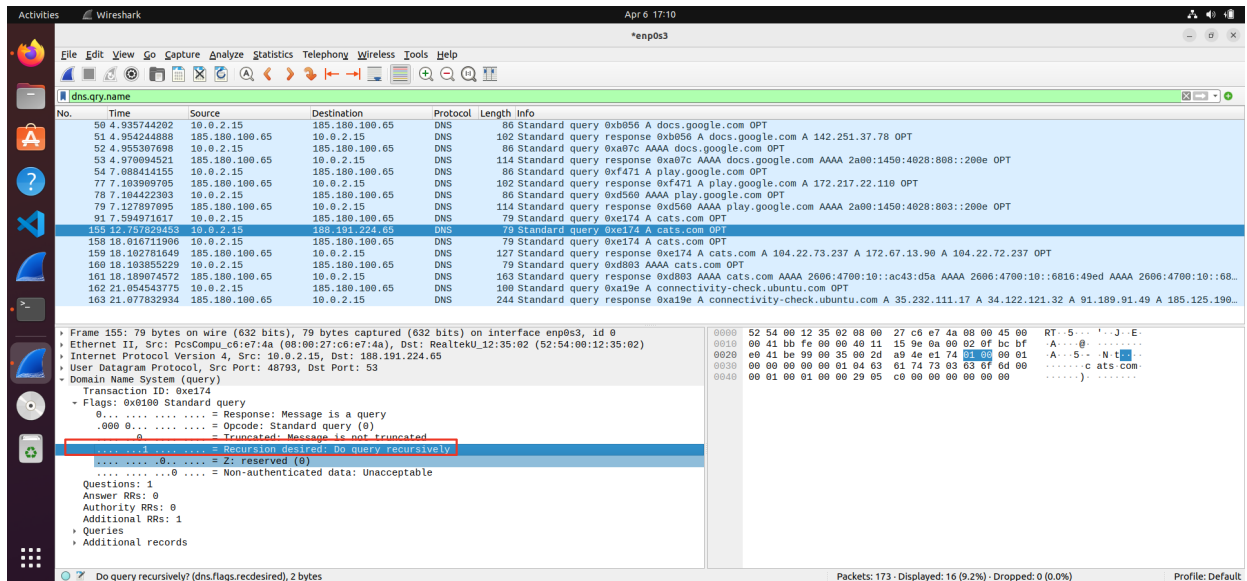
Wireshark packet capture showing DNS traffic. The packet list shows a query for 'www.google.com' from 10.0.2.15 to 185.180.100.65. The packet details pane shows the 'Destination Port (udp.port), 2 bytes' field. The packet bytes pane shows the raw data of the packet.

D:
UDP

Wireshark packet capture showing DNS traffic. The packet list shows a query for 'www.google.com' from 10.0.2.15 to 185.180.100.65. The packet details pane shows the 'UDP payload (37 bytes)' field. The packet bytes pane shows the raw data of the packet.

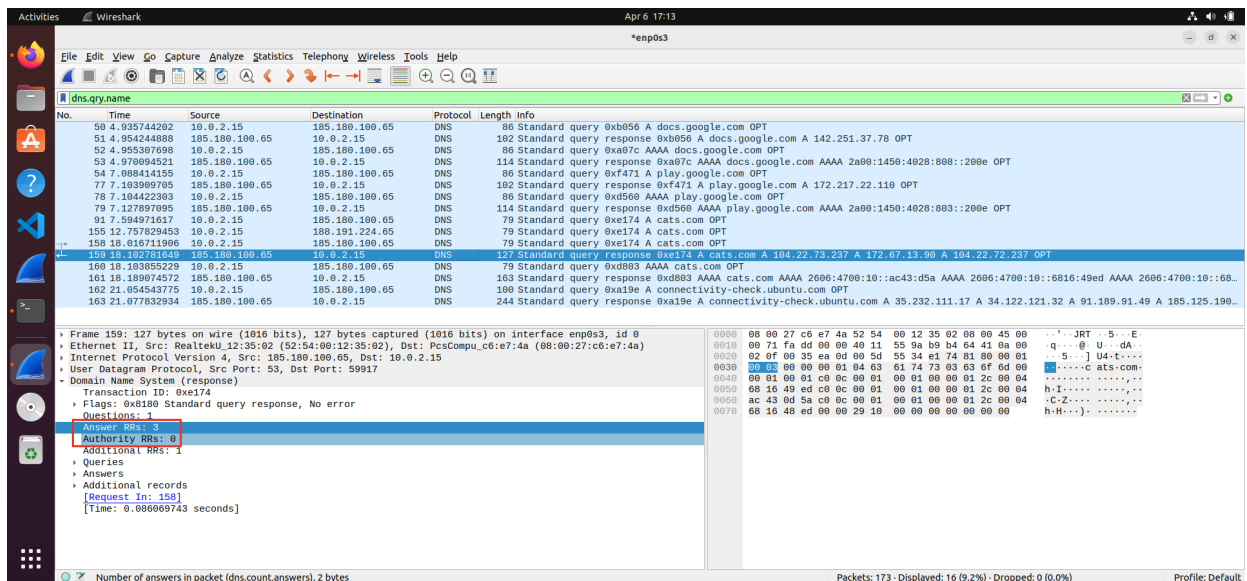
E:

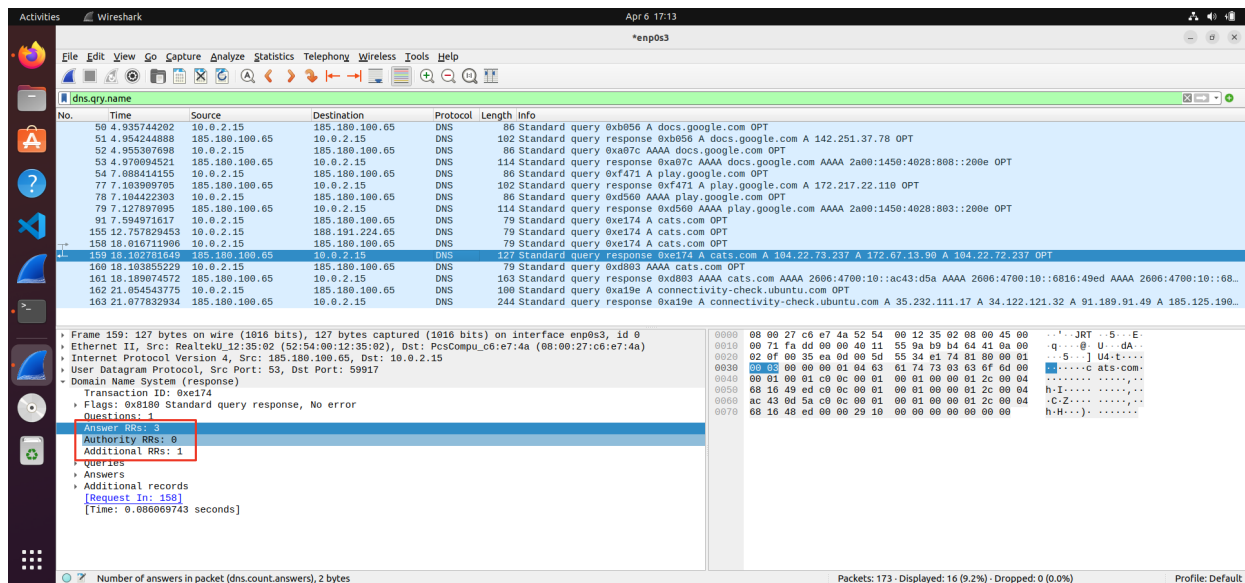
the do query recursively flag is on so it is done recursively: this can be seen by the true value of the bit in the screenshot.



F:

the first three queries received a response containing 3 answers , as did the last query response:

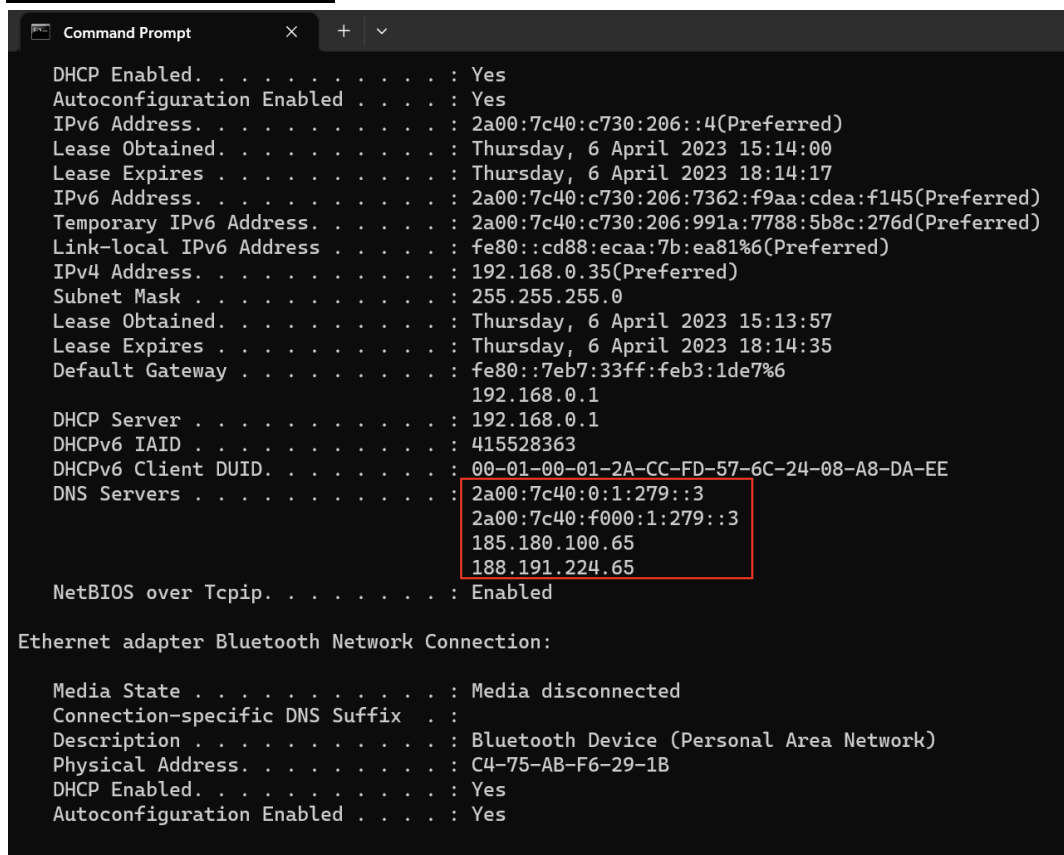




G:

A query is used to resolve a nameserver which uses IPv4. AAAA query is used to resolve a nameserver which uses IPv6.

Question number 15:



DNS server:

2a00:7c40:0:1:279::3

2a00:7c40:f000:1:279::3

185.180.100.65

188.191.224.65

Question number 16:

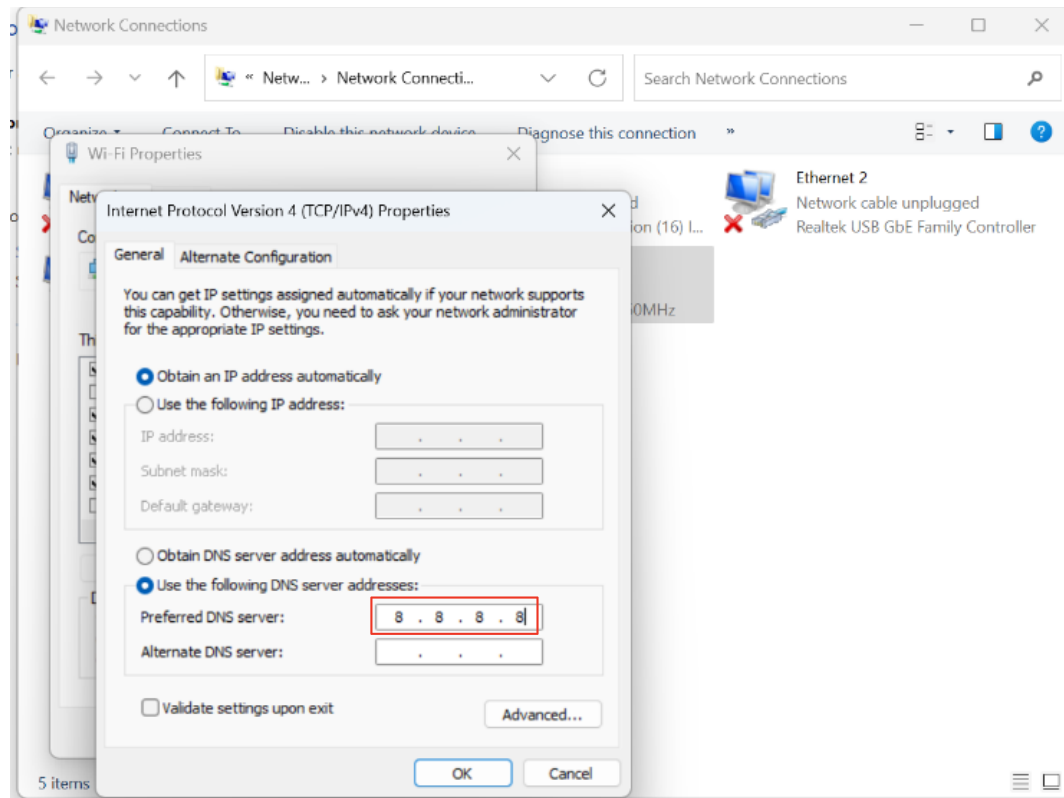
likewise in the above picture we can see:

IPv4: 192.168.0.35

IPv6: 2a00:7c40:c730:206::4

Question number 17:

properties: changing to 8.8.8.8



new DNS server:

```
192.168.0.1
DHCP Server . . . . . : 192.168.0.1
DHCPv6 IAID . . . . . : 415528363
DHCPv6 Client DUID. . . . . : 00-01-00-01-2A-CC-FD-57-6C-24-08-A8-DA-EE
DNS Servers . . . . . : 2a00:7c40:0:1:279::3
                        2a00:7c40:f000:1:279::3
                        8.8.8.8
NetBIOS over Tcpip. . . . . : Enabled
```

2a00:7c40:0:1:279::3
2a00:7c40:f000:1:279::3
8.8.8.8

Question number 18:

Without loss of generality take for an example some query to some web application say amazon. further without loss of generality let the query be “ mens black socks”. this query is a string and is quite small compared to the product listings you would receive from the response, as the response would return image of products , descriptions ect... there for the response is usually much heavier compared to the request.

Question number 19:

A:

common websites we visit on the device

```
Command Prompt
Microsoft Windows [Version 10.0.22621.1413]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Joshua>ipconfig/displaydns

Windows IP Configuration

dns.google
-----
Record Name . . . . . : dns.google
Record Type . . . . . : 28
Time To Live . . . . . : 630
Data Length . . . . . : 16
Section . . . . . : Answer
AAAA Record . . . . . : 2001:4860:4860::8888

Record Name . . . . . : dns.google
Record Type . . . . . : 28
Time To Live . . . . . : 630
Data Length . . . . . : 16
Section . . . . . : Answer
AAAA Record . . . . . : 2001:4860:4860::8844

dns.google
-----
Record Name . . . . . : dns.google
Record Type . . . . . : 1
Time To Live . . . . . : 607
Data Length . . . . . : 4
Section . . . . . : Answer
```

B:

```
C:\Users\Joshua>ipconfig/flushdns

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.

C:\Users\Joshua>
```

No DNS in cache memory as we flushed before the memory


```
Command Prompt
Microsoft Windows [Version 10.0.22621.1413]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Joshua>ipconfig/displaydns

Windows IP Configuration

C:\Users\Joshua>
```

C:

after pinging cats.com the memory of catch is now not empty and contains cats.com

```
Command Prompt

C:\Users\Joshua>ping cats.com

Pinging cats.com [2606:4700:10::6816:48ed] with 32 bytes of data:
Reply from 2606:4700:10::6816:48ed: time=92ms
Reply from 2606:4700:10::6816:48ed: time=89ms
Reply from 2606:4700:10::6816:48ed: time=94ms
Reply from 2606:4700:10::6816:48ed: time=83ms

Ping statistics for 2606:4700:10::6816:48ed:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 83ms, Maximum = 94ms, Average = 89ms

C:\Users\Joshua>ipconfig/displaydns

Windows IP Configuration

    cats.com
    -----
    Record Name . . . . . : cats.com
    Record Type . . . . . : 28
    Time To Live . . . . . : 285
    Data Length . . . . . : 16
    Section . . . . . : Answer
    AAAA Record . . . . . : 2606:4700:10::6816:48ed

    Record Name . . . . . : cats.com
    Record Type . . . . . : 28
    Time To Live . . . . . : 285
```

we can see the ip address of cats.com is and also the time to live is 285

QUESTION NUMBER 20:

The stages of surfing the internet at a website http is the following:

us as humans remember URLs and it would be really hard to remember IP addresses, therefore we type in a search engine connected to the intent of a url of a website. say for example wix.com.

a series of queries is then looked up in DNS tables to find the value of the ip address in regards to the URL. the search is first looked up in our cache and if its not there it recursively looks up higher levels until it eventually gets to a ISP . if at that stage a ip address is not found we will receive an error. if an ip address is found a GET http request is sent and we try to receive a response. if no response was given then this could mean the server is down or is having other issues. if we receive a response a connection is opened on a keepalive basis, we make a GET http request for the html file, css files and java script (as in regards to html3) where html can be described as the skeleton of the website, css the decoration of the skeleton and javascript the functionality of the skeleton. although a html.index page that is returned may have multiple files on it such as links and images the response usually "fires back all of these as well" some websites use lazy loading in order to render and reduce loading time as this can significantly hurt conversion rates and SEO as google does not reward high bounce rates and consumers have been introduced to a standard of up to 3 seconds loading. The html , css and java script are then returned and run in the web application browser . the connection is closed if there is no interaction for a predetermined amount of time.